Name
Date
Course

# Applications of Differential Calculus (Related Rates Part II) Homework

$$V=\frac{4}{3}\pi r^3$$

① A spherical balloon is inflated. At what rate is its volume increasing when its radius is 6 in and the rate that its radius is increasing is 2 in/s?

$$A=6s^2$$

② The surface area of a cube (i.e. the area of the outside surfaces of the cube) is increasing by 3 $cm^2/h$. How fast are the edge lengths of the cube increasing when the edge lengths are 10 cm?

$A = 4\pi r^2$

(3) A snowball is melting. If the surface area of the ball decreases by 3 $ft^2/min$ when the radius is 1.1 ft, at what rate is the radius of the ball decreasing?

$V = \frac{1}{3}\pi r^2 h$

(4) A water tank has the shape of a circular cone and the tip of the cone is the bottom of the tank. The radius of the top of the tank is 3 yd and the height of the tank is 9 yd. If 5 $yd^3$ of water are poured into the tank per minute, find the rate that the water level is rising when the water is 6 yd deep.

$V = \frac{1}{3}\pi r^2 h$

(5) Sand is poured onto the ground creating a pile in the shape of a circular cone. The height of the pile is always half the diameter of its base. If the diameter of its base is increasing by 3.4 m/h, what volume of sand is being poured per hour when the radius of the base of the pile is 2 m?

(6) A man 4.5 ft tall walks away from a street light at a speed of 3 ft/s. The light is 14 ft above the ground. At what rate is the length of his shadow increasing?

⑦ A light on the ground shines on a wall 16 ft away. If a women 4 ft tall walks from the light to the wall at a speed of 2 ft/s, how fast is the length of her shadow on the wall decreasing when she is 4 ft from the wall?

⑧ How fast is the tip of the shadow moving in problem #6?

Answers

1. $\boxed{288\pi \text{ in}^3/\text{s}}$ or $\boxed{904.78 \text{ in}^3/\text{s}}$
2. $\boxed{\frac{1}{40} \text{ cm/h}}$ or $\boxed{0.025 \text{ cm/h}}$
3. $\boxed{-\frac{15}{44\pi} \text{ ft/min}}$ or $\boxed{-0.1085 \text{ ft/min}}$
4. $\boxed{\frac{5}{4\pi} \text{ yd/min}}$ or $\boxed{0.3979 \text{ yd/min}}$
5. $\boxed{\frac{34\pi}{5} \text{ m}^3/\text{h}}$ or $\boxed{21.363 \text{ m}^3/\text{h}}$
6. $\boxed{\frac{27}{19} \text{ ft/s}}$ or $\boxed{1\frac{8}{19} \text{ ft/s}}$
7. $\boxed{-\frac{8}{9} \text{ ft/s}}$
8. $\boxed{\frac{84}{19} \text{ ft/s}}$ or $\boxed{4\frac{8}{19} \text{ ft/s}}$